ELITE IGCSE ACADEMY

Student Past Paper Solutions

Nov 2021 P1H Worked Solutions

Nov 2021 | P1H

21 questions ▪ question image followed by worked solution

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STUDENT EDITION

Worked answers included

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PAST PAPER QUESTION

Topic: **Expanding Brackets**

Nov 2021 P1H - Nov 2021 | P1H

Q#: 1 Marks: 3

1 (a) Simplify $e^8 \div e^2$

....

(1)

(b) Expand and simplify $(x-3)(x+1)$

....

(2)

(Total for Question 1 is 3 marks)

PAST PAPER SOLUTION

Q#: 1 Marks: 3

TOPIC: **Expanding Brackets**

Nov 2021 P1H - Nov 2021 | P1H

WORKED SOLUTION

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Topic check. Expanding brackets. The tag is correct.

Solution

(a) $e^8 \div e^2 = e^6$.

(b) $(x-3)(x+1) = x^2 - 2x - 3$.

FINAL ANSWER

(a) e^6, (b) $x^2 - 2x - 3$

PAST PAPER QUESTION

Q#: 2 Marks: 3

Topic: **Right-Angled Triangles - Pythagoras & Trigonometry**

Nov 2021 P1H - Nov 2021 | P1H

2 Here is a right-angled triangle.

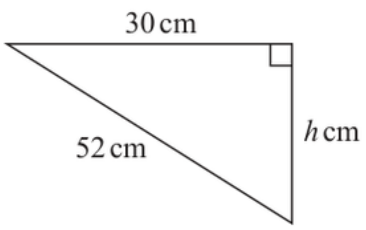

Diagram **NOT** accurately drawn

Calculate the value of h.
Give your answer correct to 3 significant figures.

h = ...

(Total for Question 2 is 3 marks)

PAST PAPER SOLUTION

Q#: 2 Marks: 3

TOPIC: **Right-Angled Triangles - Pythagoras & Trigonometry**

Nov 2021 P1H - Nov 2021 | P1H

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Topic check. Right-Angled Triangles - Pythagoras & Trigonometry. The tag is correct.

Solution

$$h^2 + 30^2 = 52^2$$

$$h^2 = 52^2 - 30^2 = 1804$$

$$h = \sqrt{1804} = 42.473\ldots$$

FINAL ANSWER

42.5 cm to 3 significant figures.

PAST PAPER QUESTION Q#: 3 Marks: 4

TOPIC: **Probability Toolkit**

Nov 2021 P1H - Nov 2021 | P1H

3 There are 54 fish in a tank.
Some of the fish are white and the rest of the fish are red.

Jeevan takes at random a fish from the tank.
The probability that he takes a white fish is $\frac{4}{9}$

(a) Work out the number of white fish originally in the tank.

....
(2)

Jeevan puts the fish he took out, back into the tank.
He puts some more white fish into the tank.

Jeevan takes at random a fish from the tank.
The probability that he takes a white fish is now $\frac{1}{2}$

(b) Work out the number of white fish Jeevan put into the tank.

....
(2)

(Total for Question 3 is 4 marks)

PAST PAPER SOLUTION

Q#: 3 Marks: 4

TOPIC: **Probability Toolkit**

Nov 2021 P1H - Nov 2021 | P1H

WORKED SOLUTION

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Topic check. Corrected to Probability Toolkit.

Solution

(a) Let the number of white fish be w.

$$\frac{w}{54} = \frac{4}{9}$$

$$w = 54 \times \frac{4}{9} = 24$$

(b) There are still 54 fish, and now

$$\frac{\text{white fish}}{54} = \frac{1}{2}$$

$$\text{white fish} = 27$$

Jeevan added

$$27 - 24 = 3$$

FINAL ANSWER

(a) 24 white fish, (b) 3 white fish.

PAST PAPER QUESTION

Q#: 4 Marks: 5

TOPIC: **Area & Perimeter**

Nov 2021 P1H - Nov 2021 | P1H

4 The diagram shows the front of a wooden door with a semicircular glass window.

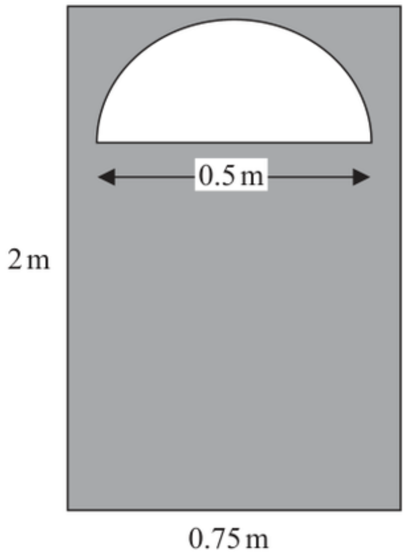

Diagram **NOT** accurately drawn

Julie wants to apply 2 coats of wood varnish to the front of the door, shown shaded in the diagram.

250 millilitres of wood varnish covers 4 m^2 of the wood.

Work out how many millilitres of wood varnish Julie will need.
Give your answer correct to the nearest millilitre.

.... millilitres

(Total for Question 4 is 5 marks)

PAST PAPER SOLUTION

Q#: 4 Marks: 5

TOPIC: **Area & Perimeter**

Nov 2021 P1H - Nov 2021 | P1H

WORKED SOLUTION

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Topic check. Corrected to Area & Perimeter. The question is about shaded area and coverage.

Solution

Area of the rectangular door:

$$0.75 \times 2 = 1.5 \text{ m}^2$$

The semicircular window has diameter 0.5 m, so

$$r = 0.25 \text{ m}$$

Area of the semicircular window:

$$\frac{1}{2}\pi(0.25)^2 = 0.098174\ldots \text{ m}^2$$

Shaded area for one coat:

$$1.5 - 0.098174\ldots = 1.401825\ldots \text{ m}^2$$

For 2 coats:

$$2 \times 1.401825\ldots = 2.80365\ldots \text{ m}^2$$

Since 250 ml covers 4 m^2, varnish needed is

$$\frac{2.80365\ldots}{4} \times 250 = 175.22\ldots$$

FINAL ANSWER

175 millilitres.

PAST PAPER QUESTION Q#: 5 Marks: 5

TOPIC: **Forming & Solving Equations**

Nov 2021 P1H - Nov 2021 | P1H

5 Yasmin has some identical rectangular tiles.
Each tile is L cm by W cm.

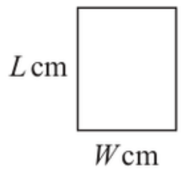

Diagram **NOT** accurately drawn

Using 9 of her tiles, Yasmin makes rectangle $ABCD$, shown in the diagram below.

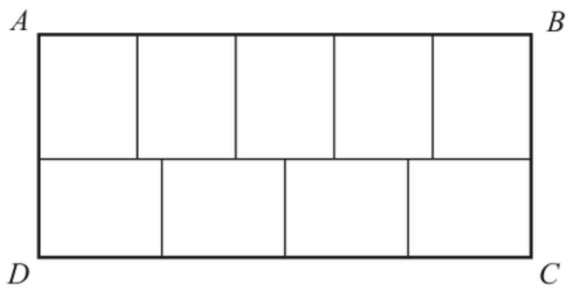

Diagram **NOT** accurately drawn

The area of $ABCD$ is $1620\,\text{cm}^2$

Work out the value of L and the value of W.

L = ... W = ...

(Total for Question 5 is 5 marks)

PAST PAPER SOLUTION

Q#: 5 Marks: 5

TOPIC: **Forming & Solving Equations**

Nov 2021 P1H - Nov 2021 | P1H

WORKED SOLUTION

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Topic check. Corrected to Forming & Solving Equations.

Solution

The top row has 5 tile widths and the bottom row has 4 tile lengths, so

$$5W = 4L$$

$$L = \frac{5}{4}W$$

The rectangle has width $5W$ and height $L + W$:

$$5W(L + W) = 1620$$

Substitute $L = \frac{5}{4}W$:

$$5W\left(\frac{5}{4}W + W\right) = 1620$$

$$5W\left(\frac{9}{4}W\right) = 1620$$

$$\frac{45}{4}W^2 = 1620$$

$$W^2 = 144$$

$$W = 12, \qquad L = \frac{5}{4}(12) = 15$$

FINAL ANSWER

$L = 15$ cm, $W = 12$ cm.

PAST PAPER QUESTION

Q#: 6 Marks: 5

TOPIC: **Simultaneous Equations**

Nov 2021 P1H - Nov 2021 | P1H

6 Alison buys 5 apples and 3 pears for a total cost of \$1.96
Greg buys 3 apples and 2 pears for a total cost of \$1.22

Michael buys 10 apples and 10 pears.

Work out how much Michael pays for his 10 apples and 10 pears.
Show your working clearly.

\$..

(Total for Question 6 is 5 marks)

PAST PAPER SOLUTION

Q#: 6 Marks: 5

TOPIC: **Simultaneous Equations**

Nov 2021 P1H - Nov 2021 | P1H

WORKED SOLUTION

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Topic check. Corrected to Simultaneous Equations.

Solution

Let a be the cost of one apple and p be the cost of one pear.

$$5a + 3p = 1.96$$

$$3a + 2p = 1.22$$

Multiply the first equation by 2:

$$10a + 6p = 3.92$$

Multiply the second equation by 3:

$$9a + 6p = 3.66$$

Subtract:

$$a = 0.26$$

Then

$$3(0.26) + 2p = 1.22$$

$$2p = 0.44$$

$$p = 0.22$$

Michael buys 10 apples and 10 pears:

$$10(0.26) + 10(0.22) = 4.80$$

FINAL ANSWER

$4.80.

PAST PAPER QUESTION

Q#: 7 Marks: 3

TOPIC: **Prime Factors, HCF & LCM**

Nov 2021 P1H - Nov 2021 | P1H

7 Write 3.6×10^3 as a product of powers of its prime factors.
Show your working clearly.

(Total for Question 7 is 3 marks)

PAST PAPER SOLUTION

Q#: 7 Marks: 3

TOPIC: **Prime Factors, HCF & LCM**

Nov 2021 P1H - Nov 2021 | P1H

WORKED SOLUTION

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Topic check. Prime factors, HCF and LCM. The tag is correct.

Method

$$3.6 \times 10^3 = 3600$$

$$3600 = 36 \times 100$$

$$36 = 2^2 \times 3^2, \qquad 100 = 2^2 \times 5^2$$

So

$$3600 = 2^4 \times 3^2 \times 5^2$$

FINAL ANSWER

$2^4 \times 3^2 \times 5^2$

PAST PAPER QUESTION

Q#: 8 Marks: 3

TOPIC: **Percentages**

Nov 2021 P1H - Nov 2021 | P1H

8 In 2018, the population of Sydney was 5.48 million.
This was 22% of the total population of Australia.

Work out the total population of Australia in 2018
Give your answer correct to 3 significant figures.

.... million

(Total for Question 8 is 3 marks)

PAST PAPER SOLUTION

Q#: 8 Marks: 3

TOPIC: **Percentages**

Nov 2021 P1H - Nov 2021 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Percentages. The tag is correct.

Method

Sydney's population was 22% of Australia's total population.

$$0.22 \times \text{total} = 5.48$$

$$\text{total} = \frac{5.48}{0.22} = 24.909\ldots$$

Correct to 3 significant figures,

$$24.909\ldots \approx 24.9$$

FINAL ANSWER
24.9 million

PAST PAPER QUESTION Q#: 9 Marks: 5

TOPIC: **Solving Inequalities**

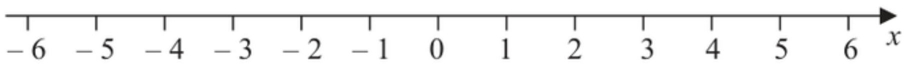
Nov 2021 P1H - Nov 2021 | P1H

9 (i) Solve the inequalities $-7 \leqslant 2x - 3 < 5$

....

(3)

(ii) On the number line, represent the solution set to part (i)



(2)

(Total for Question 9 is 5 marks)

PAST PAPER SOLUTION

Q#: 9 Marks: 5

TOPIC: **Solving Inequalities**

Nov 2021 P1H - Nov 2021 | P1H

WORKED SOLUTION

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Topic check. Solving inequalities. The tag is correct.

Solution

$$-7 \leq 2x - 3 < 5$$

Add 3 to each part:

$$-4 \leq 2x < 8$$

Divide by 2:

$$-2 \leq x < 4$$

On the number line, use a closed circle at -2, an open circle at 4, and shade between them.

FINAL ANSWER
$-2 \leq x < 4$.

PAST PAPER QUESTION Q#: 10 Marks: 5

TOPIC: **Standard & Compound Units**

Nov 2021 P1H - Nov 2021 | P1H

10 A solid aluminium cylinder has radius 10 cm and height *h* cm.

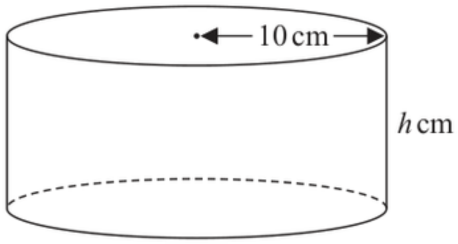

Diagram **NOT** accurately drawn

The mass of the cylinder is 5.4 kg.
The density of aluminium is 0.0027 kg/cm^3

Calculate the value of *h*.
Give your answer correct to one decimal place.

h = ...

(Total for Question 10 is 5 marks)

PAST PAPER SOLUTION

Q#: 10 Marks: 5

TOPIC: **Standard & Compound Units**

Nov 2021 P1H - Nov 2021 | P1H

WORKED SOLUTION

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Topic check. Standard & Compound Units. The tag is correct.

Solution

The mass is 5.4 kg, and the density is 0.0027 kg/cm^3.

$$\text{volume} = \frac{\text{mass}}{\text{density}} = \frac{5.4}{0.0027} = 2000 \text{ cm}^3$$

For the cylinder,

$$V = \pi r^2 h$$

$$2000 = \pi(10)^2 h$$

$$h = \frac{2000}{100\pi} = 6.366\ldots$$

FINAL ANSWER

6.4 cm.

PAST PAPER QUESTION

Q#: 11 Marks: 5

Topic: **Cumulative Frequency Diagrams**

Nov 2021 P1H - Nov 2021 | P1H

11 The table gives information about the times taken by 90 runners to complete a 10 km race.

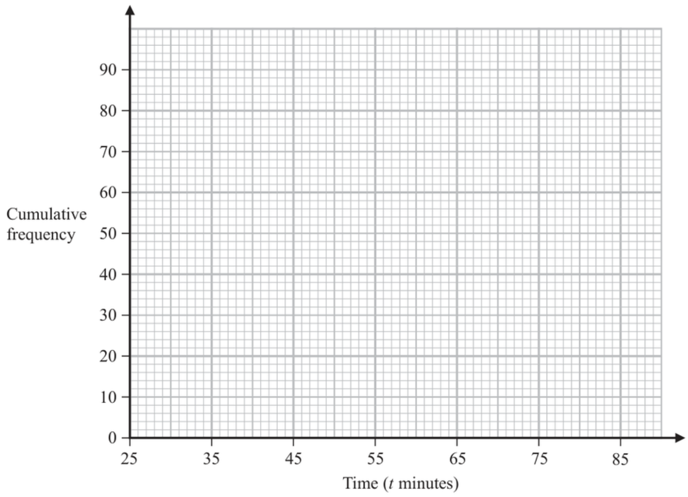
Time (*t* minutes)	**Frequency**
$25 < t \leqslant 35$	12
$35 < t \leqslant 45$	24
$45 < t \leqslant 55$	28
$55 < t \leqslant 65$	12
$65 < t \leqslant 75$	10
$75 < t \leqslant 85$	4

(a) Complete the cumulative frequency table.

Time (*t* minutes)	**Cumulative frequency**
$25 < t \leqslant 35$	12
$25 < t \leqslant 45$	
$25 < t \leqslant 55$	
$25 < t \leqslant 65$	
$25 < t \leqslant 75$	
$25 < t \leqslant 85$	

(1)

(b) On the grid below, draw a cumulative frequency graph for your table.



(2)

Any runner who completed the race in a time T minutes such that $42 < T \leqslant 52$ minutes was awarded a silver medal.

(c) Use your graph to find an estimate for the number of runners who were awarded a silver medal.

.... runners

(2)

(Total for Question 11 is 5 marks)

PAST PAPER SOLUTION

Q#: 11 Marks: 5

TOPIC: **Cumulative Frequency Diagrams**

Nov 2021 P1H - Nov 2021 | P1H

WORKED SOLUTION

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Topic check. Cumulative Frequency Diagrams. The tag is correct.

Solution

The cumulative frequencies are

$$12,\ 36,\ 64,\ 76,\ 86,\ 90$$

For $42 < T \leq 52$,

$$F(52) \approx 36 + \frac{7}{10}(64 - 36) = 55.6$$

$$F(42) \approx 12 + \frac{7}{10}(36 - 12) = 28.8$$

$$55.6 - 28.8 = 26.8$$

FINAL ANSWER

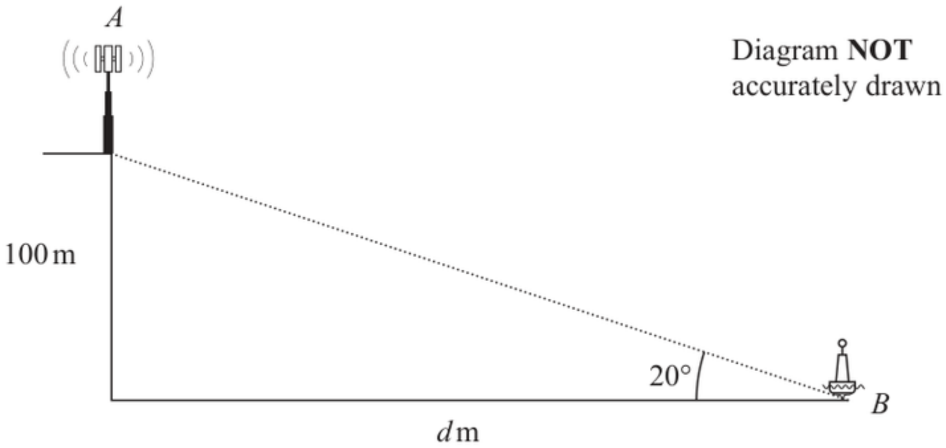
about 27 runners.

PAST PAPER QUESTION **Q#: 12** **Marks: 6**

TOPIC: **Right-Angled Triangles - Pythagoras & Trigonometry**

Nov 2021 P1H - Nov 2021 | P1H

12 The diagram shows a vertical cliff with a vertical radio mast on top of the cliff and a buoy in the sea.



The height of the cliff is 100 metres.
The buoy is at the point *B* that is *d* metres from the base of the cliff.
The angle of elevation from *B* to the top of the cliff is 20°

(a) Calculate the value of *d*.
Give your answer correct to 3 significant figures.

$d =$ ....
(3)

The point *A* at the top of the radio mast is vertically above the top of the cliff.
The angle of elevation from *B* to *A* is 25°

(b) Calculate the height of the radio mast.
Give your answer correct to 3 significant figures.

.... m
(3)

(Total for Question 12 is 6 marks)

PAST PAPER SOLUTION

Q#: 12 Marks: 6

TOPIC: **Right-Angled Triangles - Pythagoras & Trigonometry**

Nov 2021 P1H - Nov 2021 | P1H

WORKED SOLUTION

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Topic check. Right-Angled Triangles - Pythagoras & Trigonometry. The tag is correct.

Solution

(a)

$$\tan 20^\circ = \frac{100}{d}$$

$$d = \frac{100}{\tan 20^\circ} = 274.747\ldots$$

So

$$d = 275 \text{ m}$$

(b) The angle of elevation to the top of the radio mast is 25°.

$$\tan 25^\circ = \frac{100 + h}{274.747\ldots}$$

$$100 + h = 274.747\ldots \tan 25^\circ = 128.116\ldots$$

$$h = 28.116\ldots$$

FINAL ANSWER

(a) 275 m, (b) 28.1 m.

PAST PAPER QUESTION Q#: 13 Marks: 3

TOPIC: **Rounding, Estimation & Bounds**

Nov 2021 P1H - Nov 2021 | P1H

13 Here is a triangle *XYZ*.

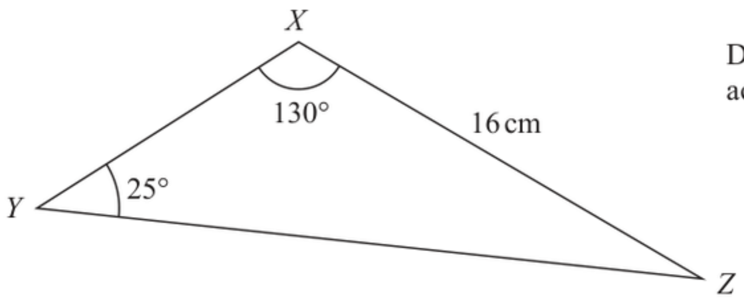

Diagram **NOT** accurately drawn

The length *XZ* and the angles *YXZ* and *XYZ* are each given correct to 2 significant figures.

Calculate the upper bound for the length *YZ*.
Give your answer correct to one decimal place.
Show your working clearly.

.... cm

(Total for Question 13 is 3 marks)

PAST PAPER SOLUTION

Q#: 13 Marks: 3

TOPIC: **Rounding, Estimation & Bounds**

Nov 2021 P1H - Nov 2021 | P1H

WORKED SOLUTION

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Topic check. Rounding, Estimation and Bounds. The tag is correct; sine rule is used inside the bounds calculation.

Method

Using the sine rule,

$$\frac{YZ}{\sin X} = \frac{XZ}{\sin Y}$$

For the upper bound of YZ:

$$XZ < 16.5, \quad X < 25.5^\circ, \quad Y < 135^\circ$$

$$YZ_{\text{upper}} = \frac{16.5 \sin 25.5^\circ}{\sin 135^\circ} = 10.0457\ldots$$

Correct to one decimal place,

$$10.0457\ldots \approx 10.0$$

FINAL ANSWER

10.0 cm

PAST PAPER QUESTION Q#: 14 Marks: 8

TOPIC: **Vectors**

Nov 2021 P1H - Nov 2021 | P1H

14 *ABCDEF* and *GHIJKL* are regular hexagons each with centre *O*.

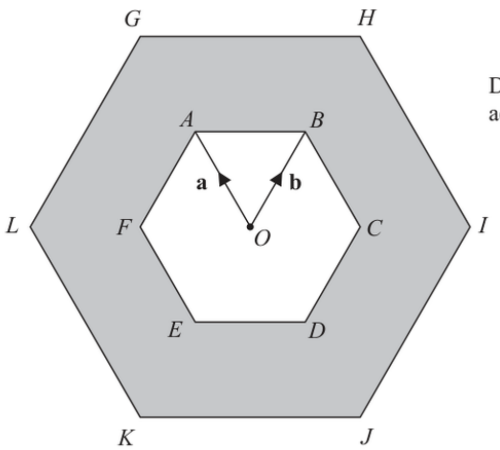

Diagram **NOT** accurately drawn

GHIJKL is an enlargement of *ABCDEF*, with centre *O* and scale factor 2

$\overrightarrow{OA} = \mathbf{a}$ $\quad \overrightarrow{OB} = \mathbf{b}$

(a) Write the following vectors, in terms of **a** and **b**.
Simplify your answers.

(i) $\overrightarrow{AB}$

....
(1)

(ii) $\overrightarrow{KI}$

....
(2)

(iii) $\overrightarrow{LD}$

....
(2)

The triangle *OAB* has an area of 5 cm^2

(b) Calculate the area of the shaded region.

.... cm^2
(3)

(Total for Question 14 is 8 marks)

PAST PAPER SOLUTION

Q#: 14 Marks: 8

TOPIC: **Vectors**

Nov 2021 P1H - Nov 2021 | P1H

WORKED SOLUTION

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Topic check. Corrected to Vectors.

Solution

$$\overrightarrow{OA} = \mathbf{a}, \qquad \overrightarrow{OB} = \mathbf{b}$$

(a)(i)

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \mathbf{b} - \mathbf{a}$$

For a regular hexagon,

$$\overrightarrow{OC} = \mathbf{b} - \mathbf{a}, \quad \overrightarrow{OD} = -\mathbf{a}, \quad \overrightarrow{OE} = -\mathbf{b}, \quad \overrightarrow{OF} = \mathbf{a} - \mathbf{b}$$

The outer hexagon has scale factor 2, so

$$\overrightarrow{OI} = 2(\mathbf{b} - \mathbf{a}), \quad \overrightarrow{OK} = -2\mathbf{b}, \quad \overrightarrow{OL} = 2(\mathbf{a} - \mathbf{b})$$

(ii)

$$\overrightarrow{KI} = \overrightarrow{OI} - \overrightarrow{OK} = 2(\mathbf{b} - \mathbf{a}) - (-2\mathbf{b}) = 4\mathbf{b} - 2\mathbf{a}$$

(iii)

$$\overrightarrow{LD} = \overrightarrow{OD} - \overrightarrow{OL} = -\mathbf{a} - 2(\mathbf{a} - \mathbf{b}) = 2\mathbf{b} - 3\mathbf{a}$$

(b) The inner hexagon is made from 6 equal triangles like OAB:

$$\text{inner area} = 6 \times 5 = 30$$

The outer hexagon is an enlargement with scale factor 2, so area scale factor is $2^2 = 4$:

$$\text{outer area} = 4 \times 30 = 120$$

$$\text{shaded area} = 120 - 30 = 90$$

FINAL ANSWER

(i) $\mathbf{b} - \mathbf{a}$, (ii) $4\mathbf{b} - 2\mathbf{a}$, (iii) $2\mathbf{b} - 3\mathbf{a}$, (b) 90 cm^2.

PAST PAPER QUESTION

Q#: 15 Marks: 8

TOPIC: **Probability Diagrams - Venn & Tree Diagrams**

Nov 2021 P1H - Nov 2021 | P1H

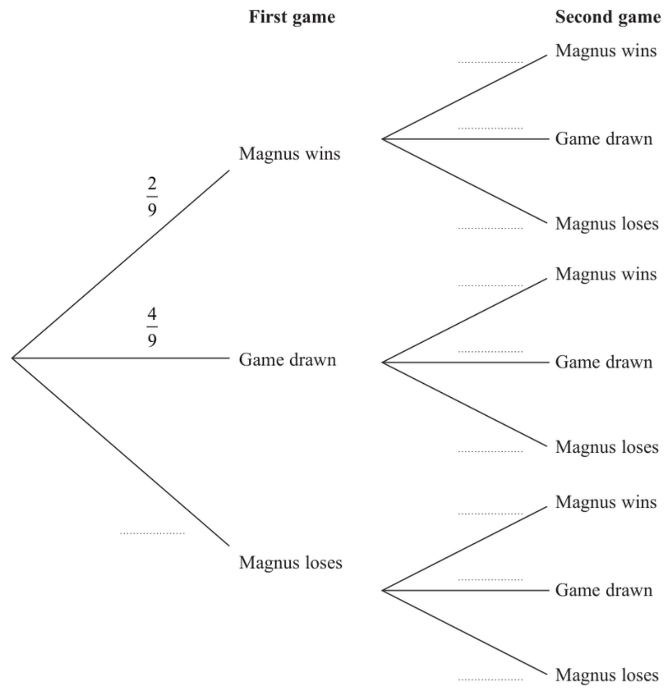
15 Magnus and Garry play 2 games of chess against each other.

The probability that Magnus beats Garry in any game is $\frac{2}{9}$

The probability that any game between Magnus and Garry is drawn is $\frac{4}{9}$

The result of any game is independent of the result of any other game.

(a) Complete the probability tree diagram.



(2)

For each game of chess,

the winner gets 2 points and the loser gets 0 points,
when the game is drawn, each player gets 1 point.

(b) Work out the probability that, after 2 games, Magnus and Garry have the same number of points.

....

(3)

Magnus and Garry now play a third game of chess.

(c) Work out the probability that, after 3 games, Magnus and Garry have the same number of points.

....

(3)

(Total for Question 15 is 8 marks)

PAST PAPER SOLUTION

Q#: 15 Marks: 8

TOPIC: **Probability Diagrams - Venn & Tree Diagrams**

Nov 2021 P1H - Nov 2021 | P1H

WORKED SOLUTION

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Topic check. Probability Diagrams - Venn & Tree Diagrams. The tag is correct.

Solution

For each chess game:

$$P(M \text{ wins}) = \frac{2}{9}, \quad P(\text{draw}) = \frac{4}{9}, \quad P(M \text{ loses}) = \frac{1}{3}$$

After two games, equal points happens if both games are drawn or Magnus wins one and loses one.

$$\left(\frac{4}{9}\right)^2 + 2\left(\frac{2}{9}\right)\left(\frac{1}{3}\right)$$

$$= \frac{16}{81} + \frac{12}{81} = \frac{28}{81}$$

After three games, equal points means Magnus scores 3 points in total.
This happens with three draws, or with one win, one draw and one loss.

$$\left(\frac{4}{9}\right)^3 + 6\left(\frac{2}{9}\right)\left(\frac{4}{9}\right)\left(\frac{1}{3}\right)$$

$$= \frac{64}{729} + \frac{144}{729} = \frac{208}{729}$$

FINAL ANSWER

$\frac{28}{81}$, $\frac{208}{729}$.

PAST PAPER QUESTION

Q#: 16 Marks: 5

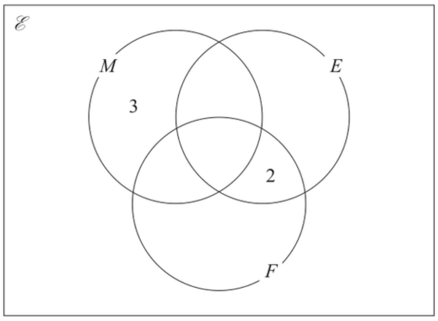
Topic: **Probability Diagrams - Venn & Tree Diagrams**

Nov 2021 P1H - Nov 2021 | P1H

16 There are 32 students in a class.
In one term these 32 students each took a test in Maths (M), in English (E) and in French (F).

25 students passed the test in Maths.
20 students passed the test in English.
14 students passed the test in French.
18 students passed the tests in Maths and English.
11 students passed the tests in Maths and French.
4 students failed all three tests.
x students passed all three tests.

The incomplete Venn diagram gives some more information about the results of the 32 students.

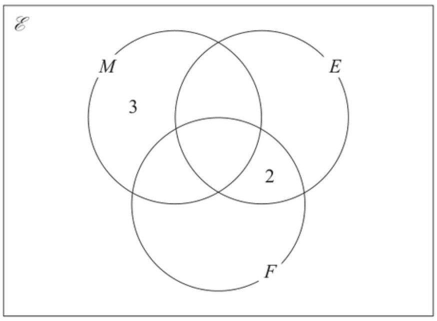


(a) Use all the given information about the results of students who passed the test in Maths to find the value of x.

x = ...

(2)

(b) Use your value of x to complete the Venn diagram to show the number of students in each subset.



(2)

A student who passed the test in Maths is chosen at random.

(c) Find the probability that this student failed the test in French.

....

(1)

(Total for Question 16 is 5 marks)

PAST PAPER SOLUTION

Q#: 16 Marks: 5

TOPIC: **Probability Diagrams - Venn & Tree Diagrams**

Nov 2021 P1H - Nov 2021 | P1H

WORKED SOLUTION

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Topic check. Probability Diagrams - Venn & Tree Diagrams. The tag is correct.

Solution

Use the Maths circle.

$$M \text{ only} = 3$$

$$M \cap E \text{ only} = 18 - x$$

$$M \cap F \text{ only} = 11 - x$$

The total in M is 25:

$$3 + (18 - x) + (11 - x) + x = 25$$

$$32 - x = 25$$

$$x = 7$$

So:

$$M \cap E \text{ only} = 11, \quad M \cap F \text{ only} = 4, \quad E \cap F \text{ only} = 2$$

$$E \text{ only} = 20 - 11 - 2 - 7 = 0$$

$$F \text{ only} = 14 - 4 - 2 - 7 = 1$$

For a student who passed Maths, failed French means M but not F:

$$\frac{3 + 11}{25} = \frac{14}{25}$$

FINAL ANSWER

$x = 7$, and $\frac{14}{25}$.

PAST PAPER QUESTION

TOPIC: **Rearranging Formulas**

Nov 2021 P1H - Nov 2021 | P1H

Q#: 17 **Marks: 8**

17 (a) Factorise $6y^2 - y - 5$

....

(2)

(b) Make f the subject of $w = \dfrac{2f + 3}{8 - f}$

....

(3)

(c) Express $4x^2 - 8x + 7$ in the form $a(x + b)^2 + c$ where a, b and c are integers.

....

(3)

(Total for Question 17 is 8 marks)

PAST PAPER SOLUTION

Q#: 17 Marks: 8

TOPIC: **Rearranging Formulas**

Nov 2021 P1H - Nov 2021 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Rearranging formulas. The tag is acceptable because rearranging is one of the main parts.

Solution

(a)

$$6y^2 - y - 5 = (6y + 5)(y - 1)$$

(b)

$$w = \frac{2f + 3}{8 - f}$$

$$w(8 - f) = 2f + 3$$

$$8w - 3 = f(w + 2)$$

$$f = \frac{8w - 3}{w + 2}$$

(c)

$$4x^2 - 8x + 7 = 4(x^2 - 2x) + 7$$

$$= 4((x - 1)^2 - 1) + 7$$

$$= 4(x - 1)^2 + 3$$

FINAL ANSWER

$(6y + 5)(y - 1)$, $f = \frac{8w-3}{w+2}$, $4(x - 1)^2 + 3$

PAST PAPER QUESTION

Q#: 18 Marks: 3

TOPIC: **Fractions, Decimals & Percentages**

Nov 2021 P1H - Nov 2021 | P1H

18 $0.4\dot{x}$ is a recurring decimal.
x is a whole number such that $1 \leqslant x \leqslant 9$

Find, in terms of x, the recurring decimal $0.4\dot{x}$ as a fraction.
Give your fraction in its simplest form.
Show clear algebraic working.

(Total for Question 18 is 3 marks)

PAST PAPER SOLUTION

Q#: 18 Marks: 3

TOPIC: **Fractions, Decimals & Percentages**

Nov 2021 P1H - Nov 2021 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Fractions, Decimals and Percentages. The tag is correct.

Method

Let

$$y = 0.4\dot{x}$$

So

$$y = 0.4xxx\ldots$$

Multiply by 10 and 100:

$$10y = 4.xxx\ldots$$

$$100y = 4x.xxx\ldots$$

Subtract:

$$100y - 10y = (4x.xxx\ldots) - (4.xxx\ldots)$$

$$90y = x + 36$$

Therefore

$$y = \frac{x+36}{90}$$

FINAL ANSWER

$\frac{x+36}{90}$

PAST PAPER QUESTION Q#: 19 Marks: 6

TOPIC: **Algebraic Proof**

Nov 2021 P1H - Nov 2021 | P1H

19 *ABCED* is a five-sided shape.

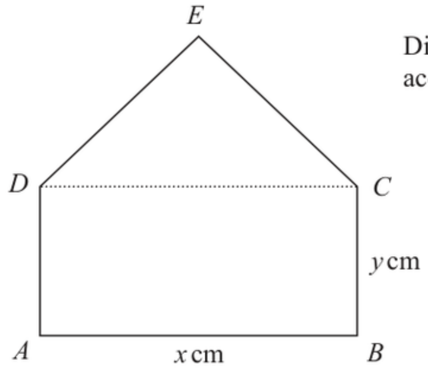

Diagram **NOT** accurately drawn

ABCD is a rectangle.
CED is an equilateral triangle.

$AB = x$ cm $\quad BC = y$ cm

The perimeter of *ABCED* is 100 cm.
The area of *ABCED* is R cm^2

(a) Show that $R = \frac{x}{4}\left(200 - \left[6 - \sqrt{3}\right]x\right)$

(3)

(b) (i) Find the value of x for which R has its maximum value.

Give your answer in the form $\frac{p}{q - \sqrt{3}}$ where p and q are integers.

$x =$ ....

(2)

(ii) Explain why the maximum value of R is given by this value of x.

....

(1)

(Total for Question 19 is 6 marks)

PAST PAPER SOLUTION

Q#: 19 Marks: 6

TOPIC: **Algebraic Proof**

Nov 2021 P1H - Nov 2021 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Algebraic proof. The tag is acceptable because the question asks to prove the formula before using it.

Solution

The perimeter is made from three sides of length x and two sides of length y:

$$3x + 2y = 100$$

So

$$y = \frac{100 - 3x}{2}$$

The area is the area of the rectangle plus the area of the equilateral triangle:

$$R = xy + \frac{\sqrt{3}}{4}x^2$$

Substitute $y = \dfrac{100 - 3x}{2}$:

$$R = x\left(\frac{100 - 3x}{2}\right) + \frac{\sqrt{3}}{4}x^2$$

$$R = 50x - \frac{3}{2}x^2 + \frac{\sqrt{3}}{4}x^2$$

$$R = \frac{x}{4}\left(200 - (6 - \sqrt{3})x\right)$$

For the maximum, write

$$R = -\frac{6 - \sqrt{3}}{4}x^2 + 50x$$

The maximum occurs at

$$x = \frac{-b}{2a} = \frac{-50}{2\left(-\frac{6-\sqrt{3}}{4}\right)}$$

$$x = \frac{100}{6 - \sqrt{3}}$$

The coefficient of x^2 is negative, so the quadratic graph opens downwards and this vertex gives a maximum.

FINAL ANSWER

$R = \dfrac{x}{4}\left(200 - (6 - \sqrt{3})x\right)$, and $x = \dfrac{100}{6 - \sqrt{3}}$.

PAST PAPER QUESTION

Q#: 20 Marks: 7

TOPIC: **Coordinate Geometry**

Nov 2021 P1H - Nov 2021 | P1H

20 The straight line **L** passes through point A (– 6, 2) and point B (5, 3)
The straight line **M** is perpendicular to **L** and passes through the midpoint of A and B.
The line **M** intersects the line $x = -1$ at point C.

Calculate the area of triangle ABC.

(Total for Question 20 is 7 marks)

PAST PAPER SOLUTION

Q#: 20 Marks: 7

TOPIC: **Coordinate Geometry**

Nov 2021 P1H - Nov 2021 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

$$A=(-6,2), \qquad B=(5,3)$$

The midpoint of AB is

$$\left(\frac{-6+5}{2},\frac{2+3}{2}\right)=\left(-\frac{1}{2},\frac{5}{2}\right)$$

The gradient of AB is

$$\frac{3-2}{5-(-6)}=\frac{1}{11}$$

So the perpendicular gradient is

$$-11$$

Line M passes through $\left(-\frac{1}{2},\frac{5}{2}\right)$, so

$$y-\frac{5}{2}=-11\left(x+\frac{1}{2}\right)$$

At point C, $x=-1$:

$$y-\frac{5}{2}=-11\left(-\frac{1}{2}\right)$$

$$y-\frac{5}{2}=\frac{11}{2}$$

$$y=8$$

So

$$C=(-1,8)$$

Now use vectors from A:

$$\vec{AB}=(11,1), \qquad \vec{AC}=(5,6)$$

Area of triangle ABC is

$$\frac{1}{2}|11(6)-1(5)|$$

$$=\frac{1}{2}(61)$$

$$=30.5$$

FINAL ANSWER

30.5

PAST PAPER QUESTION Q#: 20 Marks: 7

TOPIC: **Coordinate Geometry**

Nov 2021 P1H - Nov 2021 | P1H

20 The straight line **L** passes through point A $(-6, 2)$ and point B $(5, 3)$
The straight line **M** is perpendicular to **L** and passes through the midpoint of A and B.
The line **M** intersects the line $x = -1$ at point C.

Calculate the area of triangle ABC.

(Total for Question 20 is 7 marks)

PAST PAPER SOLUTION

Q#: 20 Marks: 7

TOPIC: **Coordinate Geometry**

Nov 2021 P1H - Nov 2021 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

$$A = (-6, 2), \qquad B = (5, 3)$$

The midpoint of AB is

$$\left(\frac{-6+5}{2}, \frac{2+3}{2}\right) = \left(-\frac{1}{2}, \frac{5}{2}\right)$$

The gradient of AB is

$$\frac{3-2}{5-(-6)} = \frac{1}{11}$$

So the perpendicular gradient is

$$-11$$

Line M passes through $\left(-\frac{1}{2}, \frac{5}{2}\right)$, so

$$y - \frac{5}{2} = -11\left(x + \frac{1}{2}\right)$$

At point C, $x = -1$:

$$y - \frac{5}{2} = -11\left(-\frac{1}{2}\right)$$

$$y - \frac{5}{2} = \frac{11}{2}$$

$$y = 8$$

So

$$C = (-1, 8)$$

Now use vectors from A:

$$\vec{AB} = (11, 1), \qquad \vec{AC} = (5, 6)$$

Area of triangle ABC is

$$\frac{1}{2}|11(6) - 1(5)|$$

$$= \frac{1}{2}(61)$$

$$= 30.5$$

FINAL ANSWER
30.5